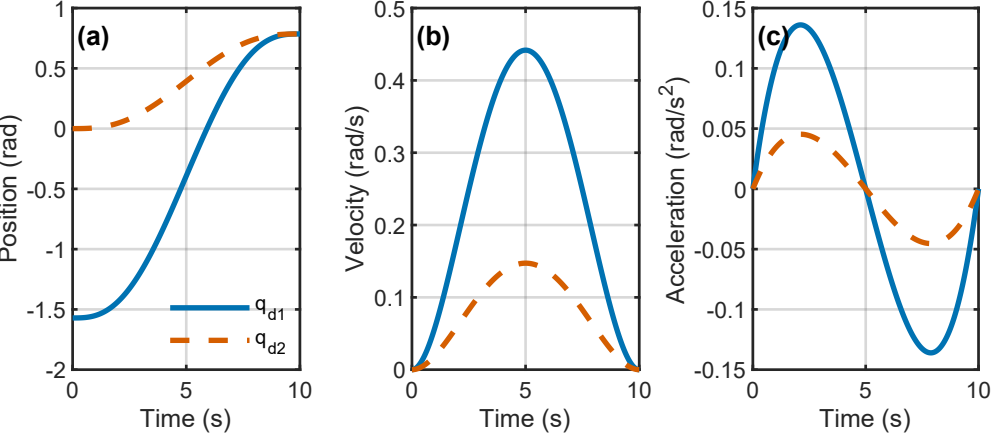

(a)
(b)
(c)
Position (rad)
Velocity (rad/s)
Acceleration (rad/s$^2$)
Time (s)
Time (s)
Time (s)
$q_{d1}$
$q_{d2}$
1
0.5
0
-0.5
-1
-1.5
-2
0.5
0.4
0.3
0.2
0.1
0
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
0
5
10